

A Survey of Copy-Number Variation Detection Tools Based on High-Throughput Sequencing Data

UNIT 7.19

Ruibin Xi,^{1,2} Semin Lee,¹ and Peter J. Park^{1,3,4}

¹Center for Biomedical Informatics, Harvard Medical School, Boston, Massachusetts

²School of Mathematical Sciences and Center for Statistical Science, Peking University, Beijing, China

³Children's Hospital Informatics Program, Boston, Massachusetts

⁴Division of Genetics, Brigham and Women's Hospital, Boston, Massachusetts

ABSTRACT

Copy-number variation (CNV) is a major class of genomic variation with potentially important functional consequences in both normal and diseased populations. Remarkable advances in development of next-generation sequencing (NGS) platforms provide an unprecedented opportunity for accurate, high-resolution characterization of CNVs. In this unit, we give an overview of available computational tools for detection of CNVs and discuss comparative advantages and disadvantages of different approaches. *Curr. Protoc. Hum. Genet.* 75:7.19.1-7.19.15. © 2012 by John Wiley & Sons, Inc.

Keywords: structural variation • insertion • deletion • indel • inversion • translocation

INTRODUCTION

Many phenotypic differences among individuals can be attributed to single-nucleotide variations (SNVs). SNVs are also associated with susceptibility or progression of various diseases, such as Crohn's disease (Barrett et al., 2008), diabetes (Steinthorsdottir et al., 2007; Barrett et al., 2009), and Parkinson's disease (Paisan-Ruiz et al., 2004), with a wide range of complexity in the number of SNVs involved and how they affect the phenotype. However, with SNVs only explaining a small fraction of variations among individuals in most cases, it is clear that other types of variations in the human genome have a significant impact on the phenotype. One class of such variations is copy-number variation (CNV), which is gain or loss of a genomic region in an individual's genome. CNVs have been shown to be common in both normal and diseased populations. In one study, as much as 12% of the genome in a healthy population has been found to contain CNVs (Redon et al., 2006); our analysis of the Database of Genomic Variants (DGV) shows that this number may be much larger. A wide spectrum of diseases such as cancer (Diskin et al., 2009; Kuiper et al., 2010; Yoshihara et al., 2011), autoimmune diseases (Fanciulli et al., 2007), schizophrenia (Stefansson et al., 2008),

and obesity (Bochukova et al., 2010) have been associated with CNVs. Thus, identification of CNVs in the human genome is essential for finding disease-related variants, both in germline and somatic samples. CNVs belong to a broader class of genomic variation referred to as structural variation, which includes both copy number-neutral changes (e.g., inversions and some translocations), as well as copy number-variant changes (e.g., duplications and deletions). In this review, we will focus on detection of copy-number-variant changes but will also briefly discuss copy-number-neutral variations.

For nearly a decade, microarray-based platforms were widely used to profile CNVs in normal genomes (Iafrate et al., 2004; Fiegler et al., 2006; Redon et al., 2006; Sebat et al., 2007; Conrad et al., 2010; Haraksingh et al., 2011), as well as in diseased genomes (Garraway et al., 2005; Zhao et al., 2005; Sebat et al., 2007; Cancer Genome Atlas Research Network, 2008; Diskin et al., 2009). These efforts have revealed that CNVs are very common and can have important functional impact. However, the resolution of microarray-based platforms is limited by the number of probes that can be placed on the array. On the latest oligonucleotide arrays, the median spacing between probes is 2 kb, making the

Searching for
Mutations

7.19.1

Supplement 75

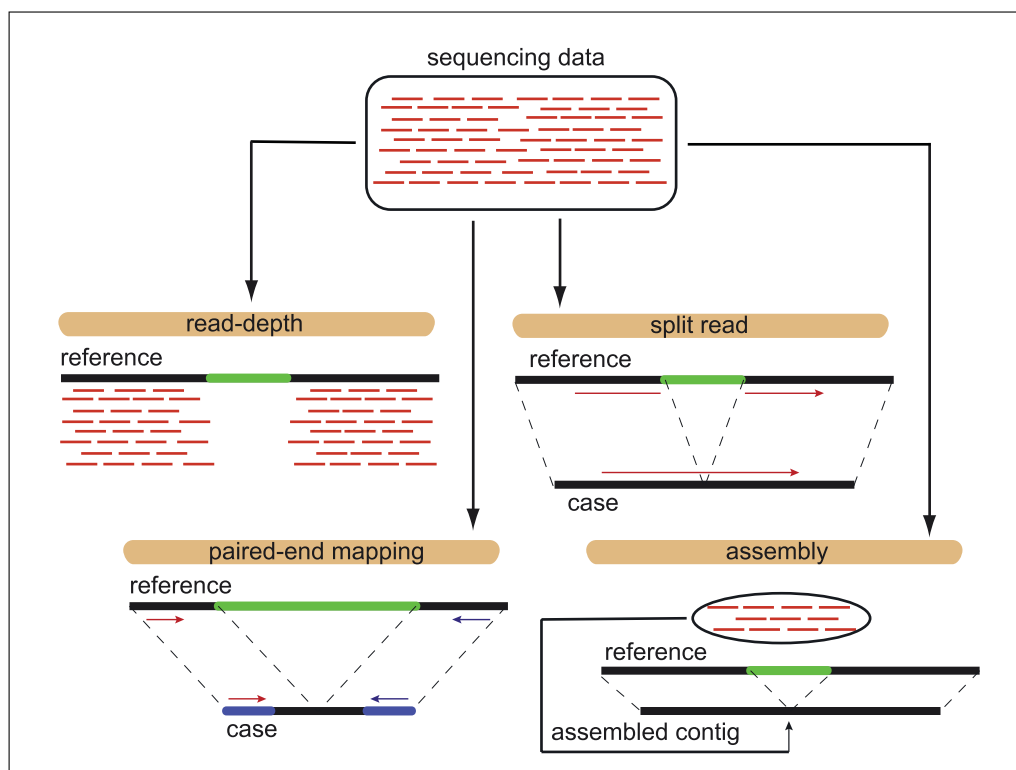


Figure 7.19.1 Four basic CNV/SV detection strategies: (1) Read-depth methods use read-density to detect CNVs. (2) Paired-end mapping (PEM) methods detect CNV/SV by analyzing configurations of PEMs. (3) Split read methods separately map the two ends of a read first to identify small insertion/deletion. Split read can also be used to pinpoint the exact location of breakpoints. (4) Assembly methods identify CNV/SV by assembling the short reads to chunks of contigs.

effective resolution for reliable calls to be ~10 to 50 kb (Pinto et al., 2011). It was not until recently that the identification of the full spectrum of CNVs became possible with the development of the next-generation sequencing (NGS) platforms (Margulies et al., 2005; Shendure et al., 2005; Bentley et al., 2008). Compared with array data, sequencing data (with a sufficient “depth” of sequencing) not only offer higher resolution, including the exact boundaries (“breakpoints”) of CNVs, but also provide the location of extra copies of DNA when there is copy number gain. To fully utilize the information provided by sequencing data, the CNV detection algorithms that we will discuss are generally more complex than those based on array platforms. An inherent challenge common to algorithms for analysis of NGS data is the large amount of data. Algorithms based on NGS data must be able to efficiently retrieve and process information from a large amount of data.

Identification of CNVs is generally done from whole-genome sequencing data. The current “high-coverage” genome typically refers to ~30× coverage of the genome (each base pair is covered 30 times on average), a somewhat arbitrary point at which one observes a

diminishing rate of return for detection of various genomic features. Depending on the feature of interest and the desired trade-off between accuracy per individual and the sampling of larger population, low-coverage (e.g., 4× to 8×) or higher-coverage (e.g., 40× to 100×) data are also available. The resolution and accuracy of CNVs and structural variations identified is directly related to the sequencing depth. The genomic coverage can be uneven throughout the genome, as some regions of the genome are not easily fragmented for sequencing and thus do not yield high coverage regardless of the overall coverage. However, for most of the genome, any discussion on the maximal resolution and accuracy must be accompanied by the sequencing depth, given that a genome can be sequenced to any desired depth.

Here, we describe available tools for detecting CNVs with NGS data. Given that CNVs constitute an aspect of structural variations (SVs), many SV detection methods involve CNV detection. Although the focus is on CNVs, we will also discuss how NGS data can be exploited to detect SVs. There are four basic strategies for detecting CNV/SV with NGS data: (1) read-depth methods, (2) paired-end

mapping (PEM) methods, (3) split read methods, and (4) assembly methods (Fig. 7.19.1). In the following, we will describe the strengths and weaknesses of each strategy in detail. We will also discuss potential issues that one should be aware of when applying these tools.

OVERVIEW OF CNV DETECTION APPROACHES BASED ON NGS DATA

The data from the NGS platforms consist of millions or even billions of short sequences (36 bp to 100 bp currently) with accompanying scores that measure the quality of each nucleotide call. The first step in most NGS data analyses is to align these short reads to a reference genome (if such a reference genome exists). In the past few years, a number of alignment tools, such as MAQ (Li, H. et al., 2008), Bowtie (Langmead et al., 2009), and BWA (Li, H. and Durbin, 2009), have been developed. Comparing with alignment tools developed for capillary reads (Ning et al., 2001; Kent, 2002), these tools are generally more efficient both in speed and in memory usage. We

refer the reader to Li and Homer (Li, H. and Homer, 2010) for a comprehensive review of these alignment tools.

Once the alignment is finished, we may apply various computational tools to identify CNVs/SVs in the sequenced genome. These computational tools are all developed based on the four aforementioned strategies. Read-depth methods are based on investigating read density along the chromosomes. A copy gain (loss) region would manifest itself with significantly more (less) mapped reads than its neighboring regions. The second approach is based on the paired-end sequencing strategy, in which both ends of each fragment are sequenced and inferences are made based on the configurations of the mapping of these paired-end reads. Split read methods and assembly methods detect CNV/SV by split mapping of short reads and assembly of short reads, respectively. In the following, we will discuss each strategy in detail.

1. Read-Depth Methods

Read-depth methods (Table 7.19.1) detect CNVs based on the density of aligned reads

Table 7.19.1 Read-Depth Methods

Algorithm	Requires control?	Availability	Reference
BIC-seq	Yes	http://compbio.med.harvard.edu/Supplements/PNAS11.html	Xi et al. (2011)
CNAseq	Yes	http://www.compbio.group.cam.ac.uk/software.html	Ivakhno et al. (2010)
CNVnator	No	http://sv.gersteinlab.org/	Abyzov et al. (2011)
CNVnorm	Yes	http://www.precancer.leeds.ac.uk/cnanorm	Gusnanto et al. (2012)
CNV-seq	Yes	http://tiger.dbs.nus.edu.sg/cnv-seq/	Xie and Tammi (2009)
CONDEX ^a	No	http://code.google.com/p/condr/	Ramachandran et al. (2011)
CONTRA ^a	YES	http://contra-cnv.sourceforge.net/	Li, J. et al. (2012)
ExomeCNV ^a	Yes	http://cran.r-project.org/web/packages/ExomeCNV/index.html	Sathirapongsasuti et al. (2011)
FREEC	Yes/No	http://bioinfo-out.curie.fr/projects/freec/	Boeva et al. (2011)
PropSeq ^a	Yes	http://bioinformatics.nki.nl/ocs/	Rigaill et al. (2012)
ReadDepth	No	http://code.google.com/p/readdepth/	Miller et al. (2011)
RDXplorer	No	http://rdxplorer.sourceforge.net/	Yoon et al. (2009)
rSW-seq	Yes	http://compbio.med.harvard.edu/Supplements/BMCBioinfo10-2.html	Kim et al. (2010)
SeqGene ^a	No	http://seqgene.sourceforge.net	Deng (2011)
SegSeq	Yes	http://www.broadinstitute.org/cgi-bin/cancer/publications/pub_paper.cgi?mode=view&paper_id=182	Chiang et al. (2009)
VarScan2 ^a	Yes	http://genome.wustl.edu/software/varscan	Koboldt et al. (2012)

^aIndicates that the algorithm is for exome-sequencing data.

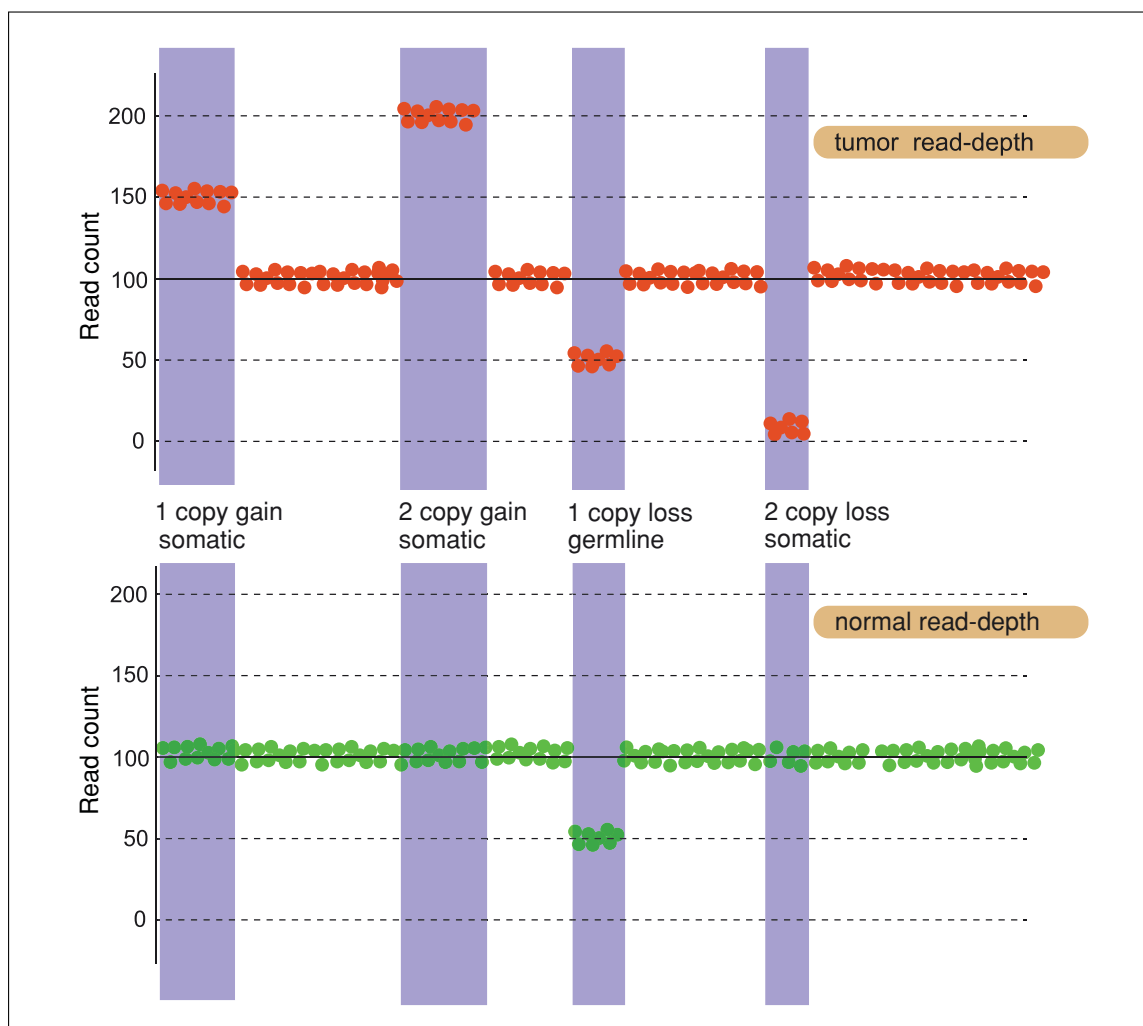


Figure 7.19.2 Read-depth methods. This approach detects CNVs by investigating the read-densities in genomic regions. Read-depth methods can be used to detect both germline and somatic CNVs.

along the chromosomes. The basic idea is that the read density of a genomic region should be roughly proportional to the copy number of the region, e.g., a deleted region should have a fewer number of mapped reads compared to CNV-free regions (Fig. 7.19.2). A complication, however is that NGS data contain various biases that can easily affect the local read density. Two major biases in the NGS data are the dependence of the read count on the GC content of the region and the variability in the mappability of short reads along the genome. The GC bias can be introduced in library construction or sequencing; depending on the experiment, read depth can be positively or negatively correlated with the GC content. The mappability bias is related to the alignment step in data processing. As the reads obtained from the current NGS platforms are short (<100 bp in most cases), a substantial portion of the reads may align to multiple positions in the reference genome, especially

in mammalian genomes with a large fraction of repetitive regions. This ambiguity depends on the read length, the aligner, and the parameters used in aligning the short reads. To avoid such ambiguity, some methods (Chiang et al., 2009; Xi et al., 2011) only consider the uniquely mapped reads, but this means that CNVs in non-uniquely mappable regions, e.g., the repetitive regions, will be undetectable. Other methods (Abyzov et al., 2011) randomly choose a position from all possible read positions given by the aligner to perform the CNV analysis. However, such random placement of the non-uniquely mapped reads could lead to many false positives.

The read-depth methods can be classified into two categories depending on whether they use a control sample or not. The methods without a control explicitly build mathematical models to account for the biases in the NGS data (mainly GC-bias). This category of methods includes RDXplorer (or EWT) (Yoon

et al., 2009), CNVnator (Abyzov et al., 2011), and FREEC (Boeva et al., 2011). The biggest advantage of the control-free methods is that they are more cost-effective as they do not require the sequencing of a control genome. The disadvantages of these control-free methods are mainly that, in addition to the known ones, there could be some unknown sources of bias and that the performance of these methods is highly dependent on how well their models can approximate the data. The other category of methods such as CNV-seq (Xie and Tammi, 2009), SegSeq (Chiang et al., 2009), rSW-seq (Kim et al., 2010), BIC-seq (Xi et al., 2011), and CNaseq (Ivakhno et al., 2010) removes biases in the NGS data by comparing a case genome with a control genome. Compared to the control-free methods, these methods can account for more sources of biases by direct subtraction of the biases from the case using the control. Methods using control data are particularly useful in cancer studies, as both the tumor and the matched normal genomes are often sequenced to distinguish somatic mutations from germline mutations. This pairing then allows one to also identify somatic CNVs by a direct comparison. Of note, the methods requiring a control genome assume that the biases in the case and the control are the same. One should make sure that the case and the control are processed in the same way, ideally by the same person, and sequenced on the same machine, before using these methods for CNV detection; otherwise, the case and the control may have different bias and these algorithms may generate false positives.

A general strategy used by many read-depth methods is to first partition the entire genome into windows such that the windows have enough mapped reads to allow a stable and accurate estimate of the read depth. For example, the CNV-seq algorithm performs a statistical test to determine an optimal window size, while EWT uses a fixed, user-defined window size (the authors of the EWT algorithm argued that window size of 100 bp is good for 30 $\times$ data). One common drawback of the window-based methods is that the resolution of breakpoint estimation cannot be finer than the window size. On the other hand, algorithms such as SegSeq, rSW-seq and BIC-seq have adopted different strategies for CNV detection. For example, BIC-seq starts with very small bins (even 1bp bins) and iteratively merges the similar neighboring bins until all neighboring bins are significantly different from each other. The choice of the bin size in this algorithm is only

for computational efficiency and thus this algorithm can achieve base-pair level resolution.

The statistical power of read-depth methods depends on the sequencing depth, the size, and the copy number of the CNV. The advantage of the read-depth methods is that they can readily detect large CNVs even with shallow read-depth. For example, Xi et al. (2011) showed that sensitivity for detecting ~ 100 kb CNVs can be $>90\%$ even at 0.3 $\times$. With deep enough sequencing data, these methods can also detect small CNVs (~ 100 bp). However, these methods will not be able to detect dosage-invariant events such as inversions and translocations.

2. Paired-End Mapping Methods

Most of the current NGS platforms allow sequencing of both ends of a DNA fragment (“paired-end reads”), in contrast to the “single-end” sequencing in which only one end of a fragment is sequenced. The paired-end sequencing technologies provide richer information than the single-end sequencing technology—the relative mapping positions of the read pairs contain important information about the structure of the sequenced genome. With the fragments in a library having a relatively uniform size, the paired-end data can be used to detect not only CNVs but also more complicated SVs. The CNV/SV detection methods based on analyzing PEMs are called PEM-based methods (Table 7.19.2).

If a paired-end read is sequenced from a genomic region without any SV, the PEM of the paired-end read should have an expected configuration, and the PEM is called a concordant PEM. The expected configuration, is dependent on the sequencing platform. For Illumina platforms, the characteristics of a concordant PEM should be (1) the mapped chromosomes are the same, (2) two ends of a read pair should be mapped to different strands, (3) the end mapped to the positive strand should be on the upstream of the other end, and (4) the distance between two mappings should be within an expected range. A PEM is called a discordant read if these conditions are not met. Figure 7.19.3 shows a set of discordant PEM configurations that can result from different classes of SVs.

The idea of PEM-based methods is to closely analyze the configurations of discordant PEMs caused by SVs and infer the likely SV. For example, if we have a PEM with two ends mapped to two chromosomes, this is indicative of an inter-chromosomal translocation in the region (assuming no sequencing

Table 7.19.2 PEM-Based Methods

Algorithm	Availability	Remark	Reference
BreakDancer	http://breakdancer.sourceforge.net/	—	Chen et al. (2009)
commonLAW	http://compbio.cs.sfu.ca/strvar.htm	Population-based	Hormozdiari et al. (2011)
GASV	http://code.google.com/p/gasv/	—	Sindi et al. (2009)
Genome STRiP	http://www.broadinstitute.org/gsa/wiki/index.php/Genome_STRiP	Population-based; only deletions	Handsaker et al. (2011)
HYDRA	http://code.google.com/p/hydra-sv/	—	Quinlan et al. (2010)
inGAP-sv	http://ingap.sourceforge.net/	—	Qi and Zhao (2011)
MoDIL	http://compbio.cs.toronto.edu/modil/	Only small Indels	Lee et al. (2009)
MoGUL	NA	Population-based	Lee et al. (2010)
PEMer	http://sv.gersteinlab.org/pemer/	—	Korbel et al. (2009)
SVDetect	http://svdetect.sourceforge.net/	—	Zeitouni et al. (2010)
VariationHunter	http://compbio.cs.sfu.ca/strvar.htm	Utilizes reads with multiple alignments	Hormozdiari et al. (2009)

or mapping error). Note that classes of discordant PEMs and classes of SVs are not one-to-one. For instance, large insertions (larger than the insert size, which is the length of the sequenced fragment) can lead to two types of discordant PEMs (Fig. 7.19.3C): one type is like a discordant PEM from a deletion (Fig. 7.19.3A), and the other like a discordant PEM from a tandem duplication (Fig. 7.19.3E). Because of such ambiguities, as well as sequencing and mapping errors, current PEM-based algorithms have limited power in detecting large insertions and more complex SVs.

The strategy of analyzing discordant PEMs was first used by Tuzun et al. (2005) and later adopted by many algorithms for CNV/SV detection. Table 7.19.2 describes some existing PEM-based algorithms. Since the insert size is only approximately known from the library generation step, the PEM-based algorithms generally derive an estimate of the insert sizes from the data. This is usually taken as the mean or the median of the mapped distances between two ends of paired-end reads. Incorporating the chromosome and orientation information, some reads can then be classified as discordant PEMs by comparing the mapping distances with the estimated insert size. The discordant PEMs with the similar configurations that support the same SV event are subsequently clustered, and SVs are inferred from these clusters of the discordant PEMs. One major difference among the PEM-based methods is how to cluster the discordant PEMs. For example, GASV (geometric analysis of structural variation) (Sindi et al., 2009) uses a geomet-

ric clustering algorithm, while BreakDancer (Chen et al., 2009) uses a window-based clustering algorithm. Similar to read-depth methods, most PEM-based algorithms only consider uniquely mapped reads, and these methods are generally not able to detect SVs in genomic regions rich with repeats and segmental duplications. An algorithm called VariationHunter (Hormozdiari et al., 2009) utilizes the multiple aligned reads and allows detection of SVs in the repeat and segmental duplication regions.

The PEM-based methods can give false positives due to various reasons. One class of the false positives is caused by complex SVs. For example, a PEM-based algorithm can predict a large insertion as a deletion because large insertions can produce deletion-like discordant PEMs. This class of false positives is generally less problematic: although the prediction of the SV type is incorrect, the breakpoint estimates of these false positives are largely correct. A second source of false-positive calls is sequencing error, which may result in a read being incorrectly mapped on the reference genome and generate a misleading discordant read. However, these sequencing errors should happen randomly and one may remove these false positives by requiring more supporting discordant PEMs. Third, chimeras can also lead to false positives. Chimera formation is not a totally random event; it is more likely to happen between DNA sequences with more homology. Since the chimeras generated before or during PCR amplification may be amplified and give many discordant PEMs, the

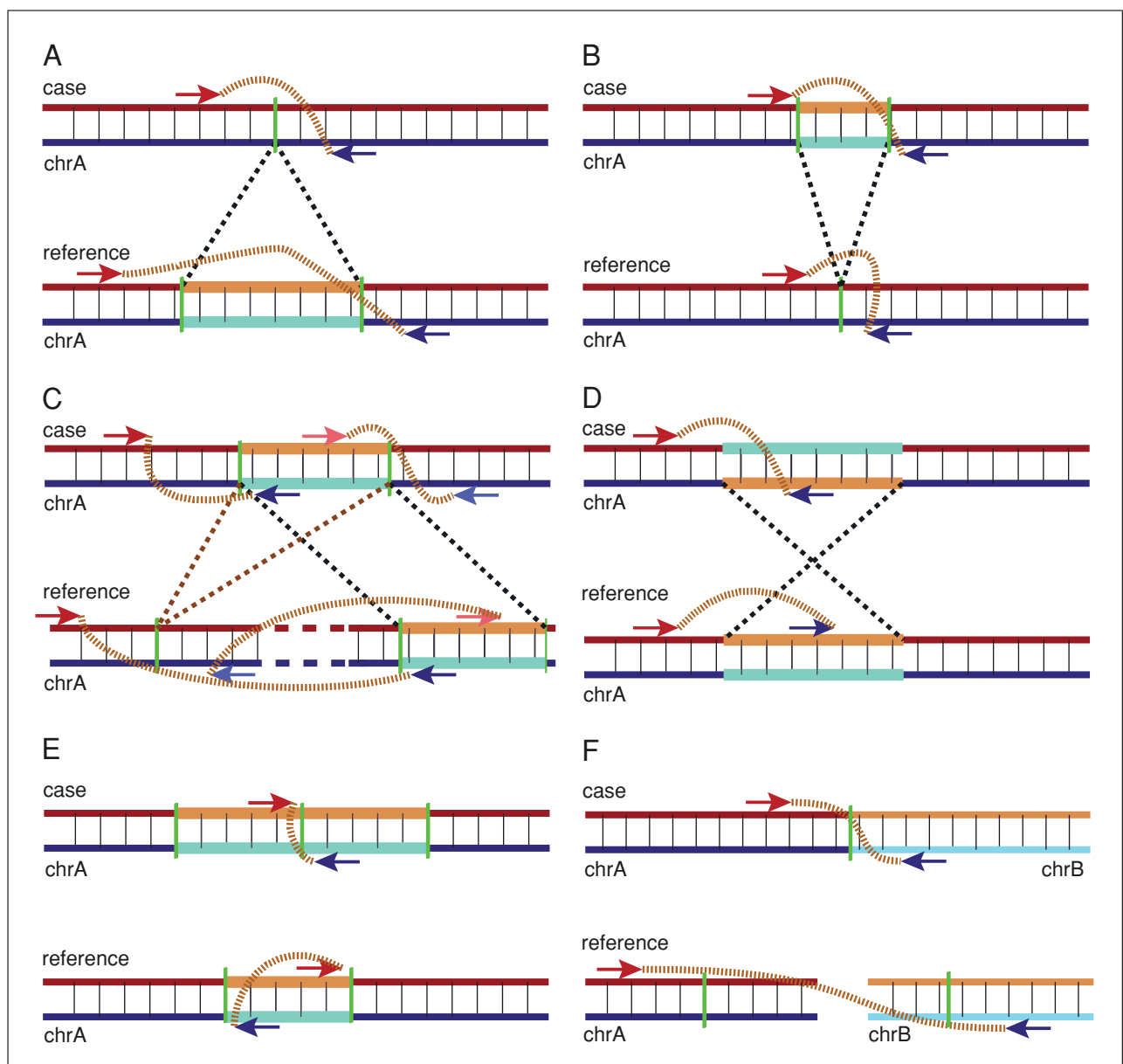


Figure 7.19.3 Configurations of discordant PEMs. (A) A fragment covering the breakpoint of a deletion is sequenced. The distance between the two ends of a read pair is significantly larger than what is expected, and hence indicates a deletion. (B) A small insertion is indicated by a read pair whose mapping distance is significantly less than what is expected. (C) A large fragment (larger than the insert size) in chromosome A is inserted into another position at chromosome A. This event will generate two classes of discordant PEMs. One class of discordant PEMs, similar to PEMs from deletions, will have mapped distances between two ends significantly larger than what is expected (the left pair on the plot). Another class of discordant PEMs will have the order of two ends switched (the right pair on the plot). (D) A read-pair mapped to the same strand signals an inversion. (E) After alignment of the read-pair shown, the mapped positions of the two ends are switched. This can be due to a tandem duplication or a large insertion (see C in this figure). (F) An inter-chromosomal can be indicated by a read-pair whose two ends are mapped to different chromosomes.

false positives from chimeras cannot be removed simply by requiring more supporting discordant PEMs. Lastly, mapping errors can contribute many false positives, which may be filtered by applying different aligners.

Due to the above artifacts, a single cluster of discordant PEMs is often not enough to signal the presence of a SV, and investigators have developed methods to further remove poten-

tial false positives. For example, the Genome STRiP algorithm (Handsaker et al., 2011) utilizes the population information and the read-depth information to perform additional filtering in deletion calls. The recent algorithm CREST (Wang et al., 2011) uses soft-clipping reads to filter false positives and refine breakpoint estimations for the BreakDancer predictions. The TIGRA package (Table 7.19.3)

Searching for Mutations

7.19.7

Table 7.19.3 Split Read and Assembly Methods

Algorithm	Availability	Remark	Reference
AGE	http://sv.gersteinlab.org/age	Split read	Abyzov et al. (2011)
CREST	http://www.stjuderesearch.org/site/lab/zhang	Soft clipping reads	Wang et al. (2011)
Nord et al.	N/A	Read-depth + split read	Nord et al. (2011)
NovelSeq	http://compbio.cs.sfu.ca/strvar.htm	PEM + assembly	Hajirasouliha et al. (2010)
Pindel	http://www.ebi.ac.uk/~kye/pindel/	Split read	Ye et al. (2009)
SLOPE	http://www-genepi.med.utah.edu/suppl/SLOPE/index.html	Split read Target Sequencing data	Abel et al. (2010)
SRiC	N/A	Split read	Zhang et al. (2011)
SVseq	http://www.engr.uconn.edu/~jjz08001/svseq.html	Split read + PEM	Zhang and Wu (2011)
TIGRA	http://genome.wustl.edu/software/tigra_sv	Assembly	N/A

instead removes false positives with a local assembly method.

3. Split Read Methods

A split read occurs when a read spans a breakpoint of a SV and thus two or more parts of the same read are mapped to different positions in the reference genome (Fig. 7.19.4). By scrutinizing reads that fail to map under a standard alignment process, split read methods (Table 7.19.3) can detect small insertions and deletions with a resolution down to a single base pair. The power of split methods, however, is heavily limited by the read length of the current technologies. For example, they can only detect insertions shorter than a read length. Considering that most of the current NGS data are only 100 bp or less, the spectrum of SVs that can be detected by split read methods is quite narrow. As the sequencing technology improves to make longer reads possible, split read methods can be adapted to detect larger and more complex SVs.

The major challenge for split read methods is the computational burden in determining where the correct split should occur. For small genomes, an otherwise unalignable read can be split at every possible position and the resulting fragments be mapped separately. Large genomes, on the other hand, contain a large fraction of repetitive regions, and short reads from each end of a fragment can potentially map to multiple locations on the reference genome. Hence, determining their best location is computationally difficult. Pindel (Ye et al., 2009) solves this problem by first an-

choring the split mapping region of a read by the mapped position of its mate and then performing the split mapping. This strategy significantly reduces the search space of split mapping and makes split mapping applicable for genomes as large as the human genome.

4. Assembly Methods

Theoretically, all genetic variations like SNVs, CNVs, and SVs can be detected with a de novo assembly method (Fig. 7.19.5). With short NGS reads, early assembly tools could only put together small bacterial genomes, but longer reads and algorithm development are gradually making it possible to assemble large genomes such as the human genome. Currently available de novo assemblers for NGS data include Velvet (Zerbino and Birney, 2008), EULER-USR (Chaisson et al., 2009), ABySS (Simpson et al., 2009), SOAPdenovo (Li et al., 2010), and ALLPATHS-LG (Gnerre et al., 2011). Compared to the assemblers developed for capillary sequencing, these tools can often perform better with short sequencing data if the coverage is sufficient. In a recent report (Li et al., 2011), the authors have demonstrated that de novo assembly of NGS reads can be used to map SVs, and 277,243 SVs ranging from 1-23 kb were identified in two human genomes.

However, de novo assembly of the entire genome based on NGS data is far from mature. Large genomes contain significant fraction of repeats and segmental duplications. For example, over 50% of DNA sequences in the human genome are repeats or segmental duplications. Current assembly algorithms based

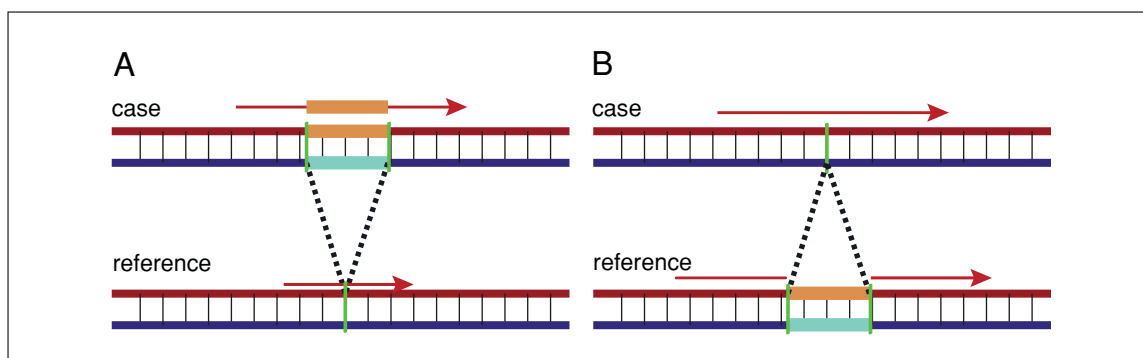


Figure 7.19.4 Split read mapping. Split read can be used to identify small insertions (A) or deletions (B).

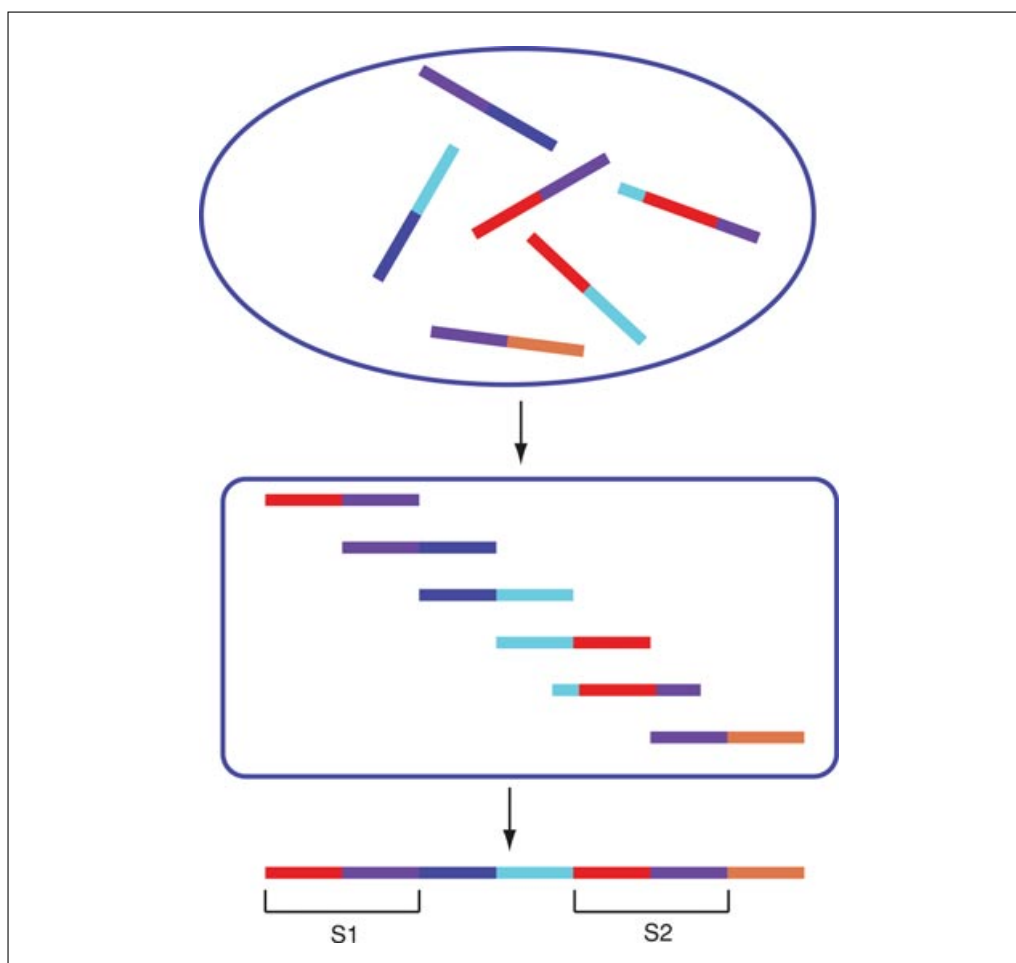


Figure 7.19.5 Assembly-based methods. In principle, assembly methods can be used to detect any genomic variations. The general strategy is to find the overlap between the short reads and assemble the short reads to contigs. In this simplified example, a copy number gain (S1 and S2 in the plot) can be identified by an assembly method.

on NGS data perform poorly in these complex regions. Although paired-end libraries with longer insert size can ameliorate this problem (Chaisson et al., 2009; Simpson et al., 2009; Li, R. et al., 2010), even the longest available inserts are insufficient to bridge many of these complex regions (Alkan et al., 2011). Another problem with the current assembly methods

is that they cannot assemble haplotype sequences and hence only homozygous structure variation can be detected (Li, Y. et al., 2011). Despite these limitations, de novo assembly are valuable for refining breakpoint estimation (the TIGRA package, see Table 7.19.3) or identifying novel insertions in the genome (Hajirasouliha et al., 2010).

Searching for Mutations

7.19.9

DISCUSSION

Although many CNV/SV detection tools for NGS data exist now, development of the computational tools has not kept up with the astonishing rate at which the sequencing technologies have improved. As a result, many genomes that are being sequenced now are not being fully analyzed for CNVs and SVs. Currently available algorithms are mostly based on only one of the four strategies as discussed above. Since the types of variants best detected are different for each strategy, an algorithm based on only one strategy has limited power. For example, a read-depth method can best detect relatively large dosage-altering CNVs, while split read methods are designed to detect small insertions and deletions. Although some algorithms have combined more than one strategy for SV detection, there is no algorithm at present that effectively integrates the four aforementioned strategies to detect the full spectrum of SVs. The Genome STRiP algorithm, for example, integrates PEM method and read-depth method for CNV detection, but this algorithm can only detect deletions and has to rely on the population information to filter false positives. Integrating the four strategies will not only broaden the detection spectrum but also increase the specificity and remove false positives.

Many factors can influence the sensitivity and specificity of CNV/SV detection. These include quality of sample preparation and library construction, the sequencing instruments, the algorithm used for CNV/SV detection, and parameters used in the algorithm. An important question to be addressed is relative contribution of these factors in CNV/SV detection and which set of protocols and parameters is optimal. For example, a larger insert size allows detection of larger insertions for a PEM method. However, libraries with a larger insert size usually also have a larger variance, which makes it difficult to detect smaller insertions and deletions. The robustness of algorithms to noisy data or low sequencing depth has not been adequately explored. This is partly due to the lack of a large database of experimentally validated SVs. Simulations are often used for benchmarking the performance of CNV/SV detection algorithms under various scenarios; although simulation studies can give some insights, they cannot provide accurate and comprehensive evaluation of the CNV/SV detection methods, as the true landscape of variants is often much more complicated than the simulated genome.

Compared with whole-genome sequencing, whole-exome sequencing has an additional step at the beginning in which fragments from the coding region are “captured.” Although the capture process does increase the overall cost, whole-exome sequencing is still substantially cheaper (e.g., 20% to 30% of the whole-genome cost currently). Because much less sequencing is required (and to compensate for the greater coverage variation, as described below), the average coverage tends to be much higher for exome data, e.g., 100× to 200× or even 500×, instead of the typical 30× to 50× for whole-genome data. This increased coverage allows for more accurate identification of SNVs or indels. However, since the capture efficiency varies considerably across regions, the coverage variation in whole-exome sequencing data is much larger than that of whole-genome sequencing data, making CNV estimation much more challenging for exome data. To account for such variation, some read-depth methods (e.g. ExomeCNV and VarScan2) for whole-exome data usually only consider regions above a specified minimum coverage. Such a region-filtering step by coverage is effective in reducing false positives in hard-to-capture regions, but it may also filter out true deletion regions. To avoid stringent filtering, a more sophisticated method may be attempted. For instance, a regression-based method to account for such variations was proposed (Rigaill et al., 2012). This method treats the read count of the case sample in each bait region (normalized by the total number of reads and the bait size) as an independent variable and models this variable as a linear function of the same statistic of the control sample. With such a regression modeling approach, this method avoids the sensitivity issue in detecting deletions found in the coverage-thresholding methods. On the other hand, by treating each bait region as a unit, the resolution of this method is limited to the bait region size. Furthermore, an important assumption in this as well as in other methods is that the biases (such as bait efficiency bias and GC-bias) in the samples are the same; a deviation from this assumption would lead to many false discoveries.

Given the complexity of the CNVs and SVs in the human genome and inadequate performance of computational algorithms at present, manual checking of the candidate cases identified by the algorithms is essential. This process would be facilitated by a visualization interface that displays a full range of data for

the user, from the raw reads mapped onto the genome to the annotation of the supportive evidence. However, even the basic steps—storing, indexing, and retrieving the relevant reads—remain challenging when working with hundreds of gigabytes per sample. A few visualization tools such as IGB (Nicol et al., 2009) and IGV (Robinson et al., 2011) for NGS data have been developed. But further development of software tools that will allow interactively visualization of both the detected SVs and the detailed NGS data would be tremendously helpful in evaluation of algorithms and in selecting the most compelling variants for further examination.

LITERATURE CITED

- Abel, H.J., Duncavage, E.J., Becker, N., Armstrong, J.R., Magrini, V.J., and Pfeifer, J.D. 2010. SLOPE: A quick and accurate method for locating non-SNP structural variation from targeted next-generation sequence data. *Bioinformatics* 26:2684-2688.
- Abyzov, A., Urban, A.E., Snyder, M., and Gerstein, M. 2011. CNVnator: An approach to discover, genotype, and characterize typical and atypical CNVs from family and population genome sequencing. *Genome Res.* 21:974-984.
- Alkan, C., Sajjadian, S., and Eichler, E.E. 2011. Limitations of next-generation genome sequence assembly. *Nat. Methods* 8:61-65.
- Barrett, J.C., Hansoul, S., Nicolae, D.L., Cho, J.H., Duerr, R.H., Rioux, J.D., Brant, S.R., Silverberg, M.S., Taylor, K.D., Barmada, M.M., Bitton, A., Dassopoulos, T., Datta, L.W., Green, T., Griffiths, A.M., Kistner, E.O., Murtha, M.T., Regueiro, M.D., Rotter, J.I., Schumm, L.P., Steinhart, A.H., Targan, S.R., Xavier, R.J., Li-biouille, C., Sandor, C., Lathrop, M., Belaiche, J., Dewit, O., Gut, I., Heath, S., Laukens, D., Mni, M., Rutgeerts, P., Van, G.A., Zelenika, D., Franchimont, D., Hugot, J.P., de, V.M., Vermeire, S., Louis, E., Cardon, L.R., Anderson, C.A., Drummond, H., Nimmo, E., Ahmad, T., Prescott, N.J., Onnie, C.M., Fisher, S.A., Marchini, J., Ghorri, J., Bumpstead, S., Gwilliam, R., Tremelling, M., Deloukas, P., Mansfield, J., Jewell, D., Satsangi, J., Mathew, C.G., Parkes, M., Georges, M., and Daly, M.J. 2008. Genome-wide association defines more than 30 distinct susceptibility loci for Crohn's Disease. *Nat. Genet.* 40:955-962.
- Barrett, J.C., Clayton, D.G., Concannon, P., Akolkar, B., Cooper, J.D., Erlich, H.A., Julier, C., Morahan, G., Nerup, J., Nierras, C., Plagnol, V., Pociot, F., Schuilenburg, H., Smyth, D.J., Stevens, H., Todd, J.A., Walker, N.M. and Rich, S.S. 2009. Genome-wide association study and meta-analysis find that over 40 loci affect risk of type 1 diabetes. *Nat. Genet.* 41:703-707.
- Bentley, D.R., Balasubramanian, S., Swerdlow, H.P., Smith, G.P., Milton, J., Brown, C.G., Hall, K.P., Evers, D.J., Barnes, C.L., Bignell, H.R., Boutell, J.M., Bryant, J., Carter, R.J., Keira, C.R., Cox, A.J., Ellis, D.J., Flatbush, M.R., Gormley, N.A., Humphray, S.J., Irving, L.J., Karbelashvili, M.S., Kirk, S.M., Li, H., Liu, X., Maisinger, K.S., Murray, L.J., Obradovic, B., Ost, T., Parkinson, M.L., Pratt, M.R., Rasolonjatovo, I.M., Reed, M.T., Rigatti, R., Rodighiero, C., Ross, M.T., Sabot, A., Sankar, S.V., Scally, A., Schroth, G.P., Smith, M.E., Smith, V.P., Spiridou, A., Torrance, P.E., Tzonev, S.S., Vermaas, E.H., Walter, K., Wu, X., Zhang, L., Alam, M.D., Anastasi, C., Aniebo, I.C., Bailey, D.M., Bancarz, I.R., Banerjee, S., Barbour, S.G., Baybayan, P.A., Benoit, V.A., Benson, K.F., Bevis, C., Black, P.J., Boodhun, A., Brennan, J.S., Bridgham, J.A., Brown, R.C., Brown, A.A., Buermann, D.H., Bundu, A.A., Burrows, J.C., Carter, N.P., Castillo, N., Chiara, E.C., Chang, S., Neil, C.R., Craike, N.R., Dada, O.O., Diakoumakos, K.D., Dominguez-Fernandez, B., Earnshaw, D.J., Egbujor, U.C., Elmore, D.W., Etchin, S.S., Ewan, M.R., Fedurco, M., Fraser, L.J., Fuentes, Fajardo, K.V., Scott, F.W., George, D., Gietzen, K.J., Goddard, C.P., Golda, G.S., Granieri, P.A., Green, D.E., Gustafson, D.L., Hansen, N.F., Harnish, K., Haudenschild, C.D., Heyer, N.I., Hims, M.M., Ho, J.T., Horgan, A.M., Hoschler, K., Hurwitz, S., Ivanov, D.V., Johnson, M.Q., James, T., Huw Jones, T.A., Kang, G.D., Kerelska, T.H., Kersey, A.D., Khrebukova, I., Kindwall, A.P., Kingsbury, Z., Kokko-Gonzales, P.I., Kumar, A., Laurent, M.A., Lawley, C.T., Lee, S.E., Lee, X., Liao, A.K., Loch, J.A., Lok, M., Luo, S., Mammen, R.M., Martin, J.W., McCauley, P.G., McNitt, P., Mehta, P., Moon, K.W., Mullens, J.W., Newington, T., Ning, Z., Ling, N.B., Novo, S.M., O'Neill, M.J., Osborne, M.A., Osnowski, A., Ostadan, O., Paraschos, L.L., Pickering, L., Pike, A.C., Pike, A.C., Chris, P.D., Pliskin, D.P., Podhasky, J., Quijano, V.J., Raczy, C., Rae, V.H., Rawlings, S.R., Chiva, R.A., Roe, P.M., Rogers, J., Rogert Bacigalupo, M.C., Romanov, N., Romieu, A., Roth, R.K., Rourke, N.J., Ruediger, S.T., Rusman, E., Sanches-Kuiper, R.M., Schenker, M.R., Seoane, J.M., Shaw, R.J., Shiver, M.K., Short, S.W., Sizto, N.L., Sluis, J.P., Smith, M.A., Ernest Sohna, S.J., Spence, E.J., Stevens, K., Sutton, N., Szajkowski, L., Tregidgo, C.L., Turcatti, G., Vandevondele, S., Verhovsky, Y., Virk, S.M., Wakelin, S., Walcott, G.C., Wang, J., Worsley, G.J., Yan, J., Yau, L., Zuerlein, M., Rogers, J., Mullikin, J.C., Hurles, M.E., McCooke, N.J., West, J.S., Oaks, F.L., Lundberg, P.L., Klenerman, D., Durbin, R., and Smith, A.J. 2008. Accurate whole human genome sequencing using reversible terminator chemistry. *Nature* 456:53-59.
- Bochukova, E.G., Huang, N., Keogh, J., Henning, E., Purmann, C., Blaszczyk, K., Saeed, S., Hamilton-Shield, J., Clayton-Smith, J., O'Rahilly, S., Hurles, M.E., and Farooqi, I.S. 2010. Large, rare chromosomal deletions associated with severe early-onset obesity. *Nature* 463:666-670.
- Boeva, V., Zinovyev, A., Bleakley, K., Vert, J.P., Janoueix-Lerosey, I., Delattre, O., and Barillot, E. 2011. Control-free calling of copy number alterations in deep-sequencing data using

- GC-content normalization. *Bioinformatics* 27:268-269.
- Cancer Genome Atlas Research Network. 2008. Comprehensive genomic characterization defines human glioblastoma genes and core pathways. *Nature* 455:1061-1068.
- Chaisson, M.J., Brinza, D., and Pevzner, P.A. 2009. De novo fragment assembly with short mate-paired reads: Does the read length matter? *Genome Res.* 19:336-346.
- Chen, K., Wallis, J.W., McLellan, M.D., Larson, D.E., Kalicki, J.M., Pohl, C.S., McGrath, S.D., Wendl, M.C., Zhang, Q., Locke, D.P., Shi, X., Fulton, R.S., Ley, T.J., Wilson, R.K., Ding, L., and Mardis, E.R. 2009. BreakDancer: An algorithm for high-resolution mapping of genomic structural variation. *Nat. Methods* 6:677-681.
- Chiang, D.Y., Getz, G., Jaffe, D.B., O'Kelly, M.J., Zhao, X., Carter, S.L., Russ, C., Nusbaum, C., Meyerson, M., and Lander, E.S. 2009. High-resolution mapping of copy-number alterations with massively parallel sequencing. *Nat. Methods* 6:99-103.
- Conrad, D.F., Pinto, D., Redon, R., Feuk, L., Gokumen, O., Zhang, Y., Aerts, J., Andrews, T.D., Barnes, C., Campbell, P., Fitzgerald, T., Hu, M., Ihm, C.H., Kristiansson, K., Macarthur, D.G., Macdonald, J.R., Onyiah, I., Pang, A.W., Robson, S., Stirrups, K., Valsesia, A., Walter, K., Wei, J., Tyler-Smith, C., Carter, N.P., Lee, C., Scherer, S.W., and Hurles, M.E. 2010. Origins and functional impact of copy number variation in the human genome. *Nature* 464:704-712.
- Deng, X. 2011. SeqGene: A comprehensive software solution for mining exome- and transcriptome-sequencing data. *BMC Bioinformatics* 12:267.
- Diskin, S.J., Hou, C., Glessner, J.T., Attiyeh, E.F., Laudenslager, M., Bosse, K., Cole, K., Mosse, Y.P., Wood, A., Lynch, J.E., Pecor, K., Diamond, M., Winter, C., Wang, K., Kim, C., Geiger, E.A., McGrady, P.W., Blakemore, A.I., London, W.B., Shaikh, T.H., Bradfield, J., Grant, S.F., Li, H., Devoto, M., Rappaport, E.R., Hakonarson, H., and Maris, J.M. 2009. Copy number variation at 1q21.1 associated with neuroblastoma. *Nature* 459:987-991.
- Fanciulli, M., Norsworthy, P.J., Petretto, E., Dong, R., Harper, L., Kamesh, L., Heward, J.M., Gough, S.C., de, S.A., Blakemore, A.I., Froguel, P., Owen, C.J., Pearce, S.H., Teixeira, L., Guillevin, L., Graham, D.S., Pusey, C.D., Cook, H.T., Vyse, T.J., and Aitman, T.J. 2007. FCGR3B copy number variation is associated with susceptibility to systemic, but not organ-specific, autoimmunity. *Nat. Genet.* 39:721-723.
- Fiegler, H., Redon, R., Andrews, D., Scott, C., Andrews, R., Carder, C., Clark, R., Dovey, O., Ellis, P., Feuk, L., French, L., Hunt, P., Kalaitzopoulos, D., Larkin, J., Montgomery, L., Perry, G.H., Plumb, B.W., Porter, K., Rigby, R.E., Rigler, D., Valsesia, A., Langford, C., Humphray, S.J., Scherer, S.W., Lee, C., Hurles, M.E., and Carter, N.P. 2006. Accurate and reliable high-throughput detection of copy number variation in the human genome. *Genome Res.* 16:1566-1574.
- Garraway, L.A., Widlund, H.R., Rubin, M.A., Getz, G., Berger, A.J., Ramaswamy, S., Beroukhi, R., Milner, D.A., Granter, S.R., Du, J., Lee, C., Wagner, S.N., Li, C., Golub, T.R., Rimm, D.L., Meyerson, M.L., Fisher, D.E., and Sellers, W.R. 2005. Integrative genomic analyses identify MITF as a lineage survival oncogene amplified in malignant melanoma. *Nature* 436:117-122.
- Gnerre, S., Maccallum, I., Przybylski, D., Ribeiro, F.J., Burton, J.N., Walker, B.J., Sharpe, T., Hall, G., Shea, T.P., Sykes, S., Berlin, A.M., Aird, D., Costello, M., Daza, R., Williams, L., Nicol, R., Gnirke, A., Nusbaum, C., Lander, E.S., and Jaffe, D.B. 2011. High-quality draft assemblies of mammalian genomes from massively parallel sequence data. *Proc. Natl. Acad. Sci. U.S.A.* 108:1513-1518.
- Gusnanto, A., Wood, H.M., Pawitan, Y., Rabbitts, P. and Berri, S. 2012. Correcting for cancer genome size and tumour cell content enables better estimation of copy number alterations from next-generation sequence data. *Bioinformatics* 28:40-47.
- Hajirasouliha, I., Hormozdiari, F., Alkan, C., Kidd, J.M., Birol, I., Eichler, E.E., and Sahinalp, S.C. 2010. Detection and characterization of novel sequence insertions using paired-end next-generation sequencing. *Bioinformatics* 26:1277-1283.
- Handsaker, R.E., Korn, J.M., Nemesh, J., and McColl, S.A. 2011. Discovery and genotyping of genome structural polymorphism by sequencing on a population scale. *Nat. Genet.* 43:269-276.
- Haraksingh, R.R., Abyzov, A., Gerstein, M., Urban, A.E., and Snyder, M. 2011. Genome-wide mapping of copy number variation in humans: Comparative analysis of high resolution array platforms. *PLoS One* 6:e27859.
- Hormozdiari, F., Alkan, C., Eichler, E.E., and Sahinalp, S.C. 2009. Combinatorial algorithms for structural variation detection in high-throughput sequenced genomes. *Genome Res.* 19:1270-1278.
- Hormozdiari, F., Hajirasouliha, I., McPherson, A., Eichler, E.E. and Sahinalp, S.C. 2011. Simultaneous structural variation discovery among multiple paired-end sequenced genomes. *Genome Res.* 21:2203-2212.
- Iafrate, A.J., Feuk, L., Rivera, M.N., Listewnik, M.L., Donahoe, P.K., Qi, Y., Scherer, S.W., and Lee, C. 2004. Detection of large-scale variation in the human genome. *Nat. Genet.* 36:949-951.
- Ivakhno, S., Royce, T., Cox, A.J., Evers, D.J., Cheetham, R.K., and Tavaré, S. 2010. CNAseg-a novel framework for identification of copy number changes in cancer from second-generation sequencing data. *Bioinformatics* 26:3051-3058.
- Kent, W.J. 2002. BLAT-the BLAST-like alignment tool. *Genome Res.* 12:656-664.
- Kim, T.M., Luquette, L.J., Xi, R., and Park, P.J. 2010. RSW-Seq: Algorithm for detection of

- copy number alterations in deep sequencing data. *BMC Bioinformatics* 11:432.
- Koboldt, D.C., Zhang, Q., Larson, D.E., Shen, D., McLellan, M.D., Lin, L., Miller, C.A., Mardis, E.R., Ding, L., and Wilson, R.K. 2012. VarScan 2: Somatic mutation and copy number alteration discovery in cancer by exome sequencing. *Genome Res.* 22:568-576.
- Korbel, J.O., Abyzov, A., Mu, X.J., Carriero, N., Cayting, P., Zhang, Z., Snyder, M. and Gerstein, M.B. 2009. PEmr: A computational framework with simulation-based error models for inferring genomic structural variants from massive paired-end sequencing data. *Genome Biol.* 10:R23.
- Kuiper, R.P., Ligtenberg, M.J., Hoogerbrugge, N., and Geurts van, K.A. 2010. Germline copy number variation and cancer risk. *Curr. Opin. Genet. Dev.* 20:282-289.
- Langmead, B., Trapnell, C., Pop, M., and Salzberg, S.L. 2009. Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol.* 10:R25.
- Lee, S., Hormozdiari, F., Alkan, C. and Brudno, M. 2009. MoDIL: Detecting small indels from clone-end sequencing with mixtures of distributions. *Nat. Methods* 6:473-474.
- Lee, S., Xing, E. and Brudno, M. 2010. MoGUL: Detecting common insertions and deletions in a population. In *Research in Computational Molecular Biology*. pp. 357-368. Springer.
- Li, H. and Durbin, R. 2009. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* 25:1754-1760.
- Li, H. and Homer, N. 2010. A survey of sequence alignment algorithms for next-generation sequencing. *Brief. Bioinform.* 11:473-483.
- Li, H., Ruan, J., and Durbin, R. 2008. Mapping short DNA sequencing reads and calling variants using mapping quality scores. *Genome Res.* 18:1851-1858.
- Li, J., Lupat, R., Amarasinghe, K.C., Thompson, E.R., Doyle, M.A., Ryland, G.L., Tothill, R.W., Halgamuge, S.K., Campbell, I.G., and Gorringe, K.L. 2012. CONTRA: Copy number analysis for targeted resequencing. *Bioinformatics* 28:1307-1313.
- Li, R., Zhu, H., Ruan, J., Qian, W., Fang, X., Shi, Z., Li, Y., Li, S., Shan, G., Kristiansen, K., Li, S., Yang, H., Wang, J., and Wang, J. 2010. De novo assembly of human genomes with massively parallel short read sequencing. *Genome Res.* 20:265-272.
- Li, Y., Zheng, H., Luo, R., Wu, H., Zhu, H., Li, R., Cao, H., Wu, B., Huang, S., Shao, H., Ma, H., Zhang, F., Feng, S., Zhang, W., Du, H., Tian, G., Li, J., Zhang, X., Li, S., Bolund, L., Kristiansen, K., de Smith, A.J., Blakemore, A.I., Coin, L.J., Yang, H., Wang, J., and Wang, J. 2011. Structural variation in two human genomes mapped at single-nucleotide resolution by whole genome de novo assembly. *Nat. Biotechnol.* 29:723-730.
- Margulies, M., Egholm, M., Altman, W.E., Attiya, S., Bader, J.S., Bembien, L.A., Berka, J., Braverman, M.S., Chen, Y.J., Chen, Z., Dewell, S.B., Du, L., Fierro, J.M., Gomes, X.V., Godwin, B.C., He, W., Helgesen, S., Ho, C.H., Irzyk, G.P., Jando, S.C., Alenquer, M.L., Jarvie, T.P., Jirage, K.B., Kim, J.B., Knight, J.R., Lanza, J.R., Leamon, J.H., Lefkowitz, S.M., Lei, M., Li, J., Lohman, K.L., Lu, H., Makhijani, V.B., McDade, K.E., McKenna, M.P., Myers, E.W., Nickerson, E., Nobile, J.R., Plant, R., Puc, B.P., Ronan, M.T., Roth, G.T., Sarkis, G.J., Simons, J.F., Simpson, J.W., Srinivasan, M., Tartaro, K.R., Tomasz, A., Vogt, K.A., Volkmer, G.A., Wang, S.H., Wang, Y., Weiner, M.P., Yu, P., Begley, R.F., and Rothberg, J.M. 2005. Genome sequencing in microfabricated high-density picolitre reactors. *Nature* 437:376-380.
- Miller, C.A., Hampton, O., Coarfa, C. and Milosavljevic, A. 2011. ReadDepth: A parallel R package for detecting copy number alterations from short sequencing reads. *PLoS One* 6:e16327.
- Nicol, J.W., Helt, G.A., Blanchard, S.G. Jr., Raja, A., and Loraine, A.E. 2009. The integrated genome browser: Free software for distribution and exploration of genome-scale datasets. *Bioinformatics* 25:2730-2731.
- Ning, Z., Cox, A.J., and Mullikin, J.C. 2001. SSAHA: A fast search method for large DNA databases. *Genome Res.* 11:1725-1729.
- Nord, A.S., Lee, M., King, M.C. and Walsh, T. 2011. Accurate and exact CNV identification from targeted high-throughput sequence data. *BMC Genomics* 12:184.
- Paisan-Ruiz, C., Jain, S., Evans, E.W., Gilks, W.P., Simon, J., van der Brug, M., Lopez de, M.A., Aparicio, S., Gil, A.M., Khan, N., Johnson, J., Martinez, J.R., Nicholl, D., Carrera, I.M., Pena, A.S., de, S.R., Lees, A., Marti-Masso, J.F., Perez-Tur, J., Wood, N.W., and Singleton, A.B. 2004. Cloning of the gene containing mutations that cause PARK8-linked Parkinson's Disease. *Neuron* 44:595-600.
- Pinto, D., Darvishi, K., Shi, X., Rajan, D., Rigler, D., Fitzgerald, T., Lionel, A.C., Thiruvahindrapuram, B., Macdonald, J.R., Mills, R., Prasad, A., Noonan, K., Gribble, S., Prigmore, E., Donahoe, P.K., Smith, R.S., Park, J.H., Hurles, M.E., Carter, N.P., Lee, C., Scherer, S.W., and Feuk, L. 2011. Comprehensive assessment of array-based platforms and calling algorithms for detection of copy number variants. *Nat. Biotechnol.* 29:512-520.
- Qi, J. and Zhao, F. 2011. inGAP-sv: A novel scheme to identify and visualize structural variation from paired end mapping data. *Nucleic Acids Res.* 39:W567-W575.
- Quinlan, A.R., Clark, R.A., Sokolova, S., Leibowitz, M.L., Zhang, Y., Hurles, M.E., Mell, J.C. and Hall, I.M. 2010. Genome-wide mapping and assembly of structural variant breakpoints in the mouse genome. *Genome Res.* 20:623-635.

- Ramachandran, A., Micsinai, M. and Pe'er, I. 2011. CONDEX: Copy number detection in exome sequences. *In* Bioinformatics and Biomedicine Workshops (BIBMW), 2011 IEEE International Conference, pp. 87-93.
- Redon, R., Ishikawa, S., Fitch, K.R., Feuk, L., Perry, G.H., Andrews, T.D., Fiegler, H., Shapero, M.H., Carson, A.R., Chen, W., Cho, E.K., Dallaire, S., Freeman, J.L., Gonzalez, J.R., Gratacos, M., Huang, J., Kalaitzopoulos, D., Komura, D., Macdonald, J.R., Marshall, C.R., Mei, R., Montgomery, L., Nishimura, K., Okamura, K., Shen, F., Somerville, M.J., Tchinda, J., Valsesia, A., Woodwark, C., Yang, F., Zhang, J., Zerjal, T., Zhang, J., Armengol, L., Conrad, D.F., Estivill, X., Tyler-Smith, C., Carter, N.P., Aburatani, H., Lee, C., Jones, K.W., Scherer, S.W. and Hurles, M.E. 2006. Global variation in copy number in the human genome. *Nature* 444:444-454.
- Rigaill, G.J., Cadot, S., Kluin, R.J.C., Xue, Z., Bernards, R., Majewski, I.J., and Wessels, L.F.A. 2012. A regression model for estimating DNA copy number data applied to capture sequencing data. *Bioinformatics* 28:2357-2365.
- Robinson, J.T., Thorvaldsdottir, H., Winckler, W., Guttman, M., Lander, E.S., Getz, G., and Mesirov, J.P. 2011. Integrative genomics viewer. *Nat. Biotechnol.* 29:24-26.
- Sathirapongsasuti, J.F., Lee, H., Horst, B.A., Brunner, G., Cochran, A.J., Binder, S., Quackenbush, J. and Nelson, S.F. 2011. Exome sequencing-based copy-number variation and loss of heterozygosity detection: ExomeCNV. *Bioinformatics* 27:2648-2654.
- Sebat, J., Lakshmi, B., Malhotra, D., Troge, J., Lese-Martin, C., Walsh, T., Yamrom, B., Yoon, S., Krasnitz, A., Kendall, J., Leotta, A., Pai, D., Zhang, R., Lee, Y.H., Hicks, J., Spence, S.J., Lee, A.T., Puura, K., Lehtimäki, T., Ledbetter, D., Gregersen, P.K., Bregman, J., Sutcliffe, J.S., Jobanputra, V., Chung, W., Warburton, D., King, M.C., Skuse, D., Geschwind, D.H., Gilliam, T.C., Ye, K., and Wigler, M. 2007. Strong association of de novo copy number mutations with autism. *Science* 316:445-449.
- Shendure, J., Porreca, G.J., Reppas, N.B., Lin, X., McCutcheon, J.P., Rosenbaum, A.M., Wang, M.D., Zhang, K., Mitra, R.D., and Church, G.M. 2005. Accurate multiplex polony sequencing of an evolved bacterial genome. *Science* 309:1728-1732.
- Simpson, J.T., Wong, K., Jackman, S.D., Schein, J.E., Jones, S.J., and Birol, I. 2009. ABySS: A parallel assembler for short read sequence data. *Genome Res.* 19:1117-1123.
- Sindi, S., Helman, E., Bashir, A., and Raphael, B.J. 2009. A geometric approach for classification and comparison of structural variants. *Bioinformatics* 25:i222-i230.
- Stefansson, H., Rujescu, D., Cichon, S., Pietiläinen, O.P., Ingason, A., Steinberg, S., Fossdal, R., Sigurdsson, E., Sigmundsson, T., Buizer-Voskamp, J.E., Hansen, T., Jakobsen, K.D., Muglia, P., Francks, C., Matthews, P.M., Gylfason, A., Halldorsson, B.V., Gudbjartsson, D., Thorgeirsson, T.E., Sigurdsson, A., Jonasdottir, A., Jonasdottir, A., Bjornsson, A., Mattiasdottir, S., Blondal, T., Haraldsson, M., Magnusdottir, B.B., Giegling, I., Moller, H.J., Hartmann, A., Shanna, K.V., Ge, D., Need, A.C., Crombie, C., Fraser, G., Walker, N., Lonnqvist, J., Suvisaari, J., Tuulio-Henriksson, A., Paunio, T., Toupoulou, T., Bramon, E., Di, F.M., Murray, R., Ruggeri, M., Vassos, E., Tosato, S., Walshe, M., Li, T., Vasilescu, C., Muhleisen, T.W., Wang, A.G., Ullum, H., Djurovic, S., Melle, I., Olesen, J., Kiemeny, L.A., Franke, B., Sabatti, C., Freimer, N.B., Gulcher, J.R., Thorsteinsdottir, U., Kong, A., Andreassen, O.A., Ophoff, R.A., Georgi, A., Rietschel, M., Werge, T., Petursson, H., Goldstein, D.B., Nothen, M.M., Peltonen, L., Collier, D.A., St, C.D. and Stefansson, K. 2008. Large recurrent microdeletions associated with schizophrenia. *Nature* 455:232-236.
- Steinthorsdottir, V., Thorleifsson, G., Reynisdottir, I., Benediktsson, R., Jonsdottir, S., Walters, G.B., Styrkarsdottir, U., Gretarsdottir, S., Emilsson, V., Ghosh, S., Baker, A., Snorraddottir, S., Bjarnason, H., Ng, M.C., Hansen, T., Bagger, Y., Wilensky, R.L., Reilly, M.P., Adeyemo, A., Chen, Y., Zhou, J., Gudnason, V., Chen, G., Huang, H., Lashley, K., Doumatey, A., So, W.Y., Ma, R.C., Andersen, G., Borch-Johnsen, K., Jorgensen, T., van Vliet-Ostapchouk, J.V., Hofker, M.H., Wijmenga, C., Christiansen, C., Rader, D.J., Rotimi, C., Gurney, M., Chan, J.C., Pedersen, O., Sigurdsson, G., Gulcher, J.R., Thorsteinsdottir, U., Kong, A., and Stefansson, K. 2007. A variant in CDKAL1 influences insulin response and risk of Type 2 Diabetes. *Nat. Genet.* 39:770-775.
- Tuzun, E., Sharp, A.J., Bailey, J.A., Kaul, R., Morrison, V.A., Pertz, L.M., Haugen, E., Hayden, H., Albertson, D., Pinkel, D., Olson, M.V., and Eichler, E.E. 2005. Fine-scale structural variation of the human genome. *Nat. Genet.* 37:727-732.
- Wang, J., Mullighan, C.G., Easton, J., Roberts, S., Heatley, S.L., Ma, J., Rusch, M.C., Chen, K., Harris, C.C., Ding, L., Holmfeldt, L., Payne-Turner, D., Fan, X., Wei, L., Zhao, D., Obenaus, J.C., Naeve, C., Mardis, E.R., Wilson, R.K., Downing, J.R., and Zhang, J. 2011. CREST maps somatic structural variation in cancer genomes with base-pair resolution. *Nat. Methods* 8:652-654.
- Xi, R., Hadjipanayis, A.G., Luquette, L.J., Kim, T.M., Lee, E., Zhang, J., Johnson, M.D., Muzny, D.M., Wheeler, D.A., Gibbs, R.A., Kucherlapati, R., and Park, P.J. 2011. Copy number variation detection in whole-genome sequencing data using the Bayesian information criterion. *Proc. Natl. Acad. Sci. U.S.A.* 108:E1128-E1136.
- Xie, C. and Tammi, M.T. 2009. CNV-Seq, a new method to detect copy number variation using high-throughput sequencing. *BMC Bioinformatics* 10:80.
- Ye, K., Schulz, M.H., Long, Q., Apweiler, R., and Ning, Z. 2009. Pindel: A pattern growth approach to detect break points of large deletions and medium sized insertions from paired-end short reads. *Bioinformatics* 25:2865-2871.

- Yoon, S., Xuan, Z., Makarov, V., Ye, K., and Sebat, J. 2009. Sensitive and accurate detection of copy number variants using read depth of coverage. *Genome Res.* 19:1586-1592.
- Yoshihara, K., Tajima, A., Adachi, S., Quan, J., Sekine, M., Kase, H., Yahata, T., Inoue, I., and Tanaka, K. 2011. Germline copy number variations in BRCA1-associated ovarian cancer patients. *Genes Chromosomes Cancer* 50:167-177.
- Zeitouni, B., Boeva, V., Janoueix-Lerosey, I., Loeillet, S., Legoix-né, P., Nicolas, A., Delattre, O., and Barillot, E. 2010. SVDetect: A tool to identify genomic structural variations from paired-end and mate-pair sequencing data. *Bioinformatics* 26:1895-1896.
- Zerbino, D.R. and Birney, E. 2008. Velvet: Algorithms for de novo short read assembly using de bruijn graphs. *Genome Res.* 18:821-829.
- Zhang, J. and Wu, Y. 2011. SVseq: An approach for detecting exact breakpoints of deletions with low-coverage sequence data. *Bioinformatics* 27:3228-3234.
- Zhang, Z.D., Du, J., Lam, H., Abyzov, A., Urban, A.E., Snyder, M., and Gerstein, M. 2011. Identification of genomic indels and structural variations using split reads. *BMC Genomics* 12:375.
- Zhao, X., Weir, B.A., LaFramboise, T., Lin, M., Beroukheim, R., Garraway, L., Beheshti, J., Lee, J.C., Naoki, K., Richards, W.G., Sugarbaker, D., Chen, F., Rubin, M.A., Janne, P.A., Girard, L., Minna, J., Christiani, D., Li, C., Sellers, W.R., and Meyerson, M. 2005. Homozygous deletions and chromosome amplifications in human lung carcinomas revealed by single nucleotide polymorphism array analysis. *Cancer Res.* 65:5561-5570.